

Block-Power Quotient Envelope Criterion

Bin Wang

July 2026

Abstract

The orthogonal quotient of RH-245 has no one-step contraction, so a direct geometric estimate is unavailable. We prove the exact block-power substitute: a uniformly bounded trace norm and a uniformly contractive operator norm for one power C^m imply a geometric envelope for all later quotient traces and an explicit logarithmic tail bound. A finite $m = 12$ diagnostic on 17 tractable endpoints gives rate 0.393300 and unit-disk tail bound 1.80×10^{-5} . The constants are not promoted to a noise-uniform certificate.

1 Block criterion

Let $C_\sigma \in \mathcal{S}_2(H)$ be the orthogonal quotient from RH-245 and $\tau_n(\sigma) = \text{Tr } C_\sigma^n$ for $n \geq 2$ [2]. Fix an integer $m \geq 2$. Suppose

$$K_m = \sup_\sigma \|C_\sigma^m\|_1 < \infty, \quad \eta_m = \sup_\sigma \|C_\sigma^m\| < 1,$$

and set $L_r = \sup_\sigma \|C_\sigma^r\| < \infty$ for $0 \leq r < m$ (with $L_0 = 1$).

Theorem 1 (block-power envelope). *For $n = \ell m + r \geq m$, $\ell \geq 1$, $0 \leq r < m$,*

$$|\tau_n(\sigma)| \leq K_m L_r \eta_m^{\ell-1}. \quad (1)$$

Writing $q = \eta_m^{1/m}$, the right side is at most $M_m q^n$, where

$$M_m = K_m \max_{0 \leq r < m} L_r q^{-(m+r)}. \quad (2)$$

If the finite head $2 \leq n < m$ has a uniform bound compatible with the same rate, then $|\tau_n(\sigma)| \leq M q^n$ for every $n \geq 2$ and RH-240 applies.

Proof. Factor $C_\sigma^n = (C_\sigma^m)^\ell C_\sigma^r$ and use $|\text{Tr } T| \leq \|T\|_1$, submultiplicativity, and the ideal inequality $\|UV\|_1 \leq \|U\|_1 \|V\|$. This gives (1). Since $\eta_m^{\ell-1} = q^{n-(m+r)}$, (2) follows. The finite head is handled by enlarging M finitely; RH-240 then supplies local normality and zero-freeness of the relative determinant [1]. $\square$

The same factorization yields a useful tail estimate. For $R \geq 0$ with $\eta_m R^m < 1$,

$$\sum_{n \geq m} \frac{|\tau_n(\sigma)| R^n}{n} \leq \frac{K_m R^m}{m(1 - \eta_m R^m)} \sum_{r=0}^{m-1} L_r R^r. \quad (3)$$

This is a certificate template, not a claim that its hypotheses hold uniformly for the moving cloud.

2 Finite 12-block diagnostic

We use the 17 ordered-Schur quotient rows of RH-245 (dimensions at most 512) and take maxima over this finite subbatch. The first contractive depth is 3–7, with no one-step contraction. For $m = 12$,

$$\eta_{12} \leq 1.3698766308677744 \times 10^{-5}, \quad K_{12} \leq 1.5684781027206807 \times 10^{-5},$$

so

$$q_{12} = 0.3932995547481413, \quad \sum_{n \geq 12} \frac{|\tau_n|}{n} \leq 1.7991531976413385 \times 10^{-5}$$

under the finite-sample constants and $R = 1$. The corresponding finite remainder operator-norm maxima for orders 0 through 11 are retained in the JSON audit, and give $M_{12} = 697.6520352674352$ through (2). The large M is a reminder that a fast asymptotic block rate does not by itself make the first uncontrolled orders small.

quantity	value
finite endpoints	17
first contractive depth	3–7
one-step contractive endpoints	0/17
q_{12}	0.3932995547
M_{12}	697.6520353
unit-disk tail from order 12	1.7992×10^{-5}

3 Boundary

The theorem is an exact conditional route to the all-order envelope. The finite diagnostic does not establish uniform K_m , η_m , or L_r over all 32 endpoints, interval perturbations, or a continuum of noise. It also does not identify a cloud whose residual has the RH-243 deterministic anchor. The separate-absolute route is tested next; it is expected to fail because it discards the cancellation that makes the quotient small.

Gate A remains open and Gates B–E are untouched. No Hilbert–Pólya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

- [1] Bin Wang. Uniform trace envelope criterion. 2026. RH-240 preprint.
- [2] Bin Wang. Orthogonal-quotient superloop compression. 2026. RH-245 preprint.